

Lecture 7: Ito Calculus & Ito's Lemma

Leili Rafiee Sevyeri

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Cheriton School of Computer Science
University of Waterloo

Back to Ito Calculus

Suppose we have a variable X undergoing Brownian motion (as usual),

$$dX = a(X, t)dt + b(X, t)dZ \quad \text{with} \quad dZ = \phi\sqrt{dt}, \quad \phi \sim N(0, 1)$$

and F is a function depending on X and t , i.e., $F(X(t), t)$.

How should F evolve?

As $t \rightarrow t + dt$ and $F \rightarrow F + dF$, then a Taylor series implies that...

$$dF = \frac{\partial F}{\partial t} dt + \frac{\partial F}{\partial X} dX + \frac{1}{2} \frac{\partial^2 F}{\partial X^2} (dX)^2 + \dots$$

Let's work out dX^2 and apply Big- O to simplify!

Deriving Ito's Lemma

We can multiply through to find that:

$$(dX)^2 = (adt + bdZ)^2 = a^2 dt^2 + 2ab dt \cdot dZ + b^2 (dZ)^2$$

and we remember that:

$$dZ = \phi \sqrt{dt} = O(dt^{0.5}) \quad \text{as } dt \rightarrow 0$$

$$(dZ)^2 = dt \quad \text{as } dt \rightarrow 0 \quad \leftarrow \text{from slide 9}$$

Therefore $(dX)^2 = a^2 dt^2 + 2ab \times O(dt^{1.5}) + b^2 dt$

Or, again applying Big-O (assuming small dt),

$$(dX)^2 = b^2 dt + O(dt^{1.5})$$

Now let's go back to our expression for dF ...

Deriving Ito's Lemma

Now:

$$\begin{aligned} dF &= F_t dt + F_x dX + \frac{1}{2} F_{xx} (dX)^2 + \dots \\ &= F_t dt + F_x (adt + bdZ) + \frac{1}{2} F_{xx} (b^2 dt) + O(dt)^{1.5} \end{aligned}$$

Grouping terms:

$$= bF_x dZ + \left(aF_x + \frac{b^2}{2} F_{xx} + F_t \right) dt + O(dt)^{1.5}$$

We again have a drift term and a random term, but also one more that is “small” (i.e., goes to zero).

End of Lecture 6

A note on the last (“small”) term

Remember we interpreted our SDEs as describing integrals, so for dF we have:

$$\int_0^t dF = \int_0^t \left(bF_X dZ + \left(aF_X + \frac{b^2}{2} F_{XX} + F_t \right) dt \right) + O \left(\underbrace{\int_0^t dt^{1.5}}_0 \right)$$

Consider the last term as the limit of a discrete sum: $N \cdot \Delta t$

$$\begin{aligned} \int_0^t dt^{1.5} &= \lim_{\Delta t \rightarrow 0} \sum_0^{N-1} (\Delta t)^{1.5} = \lim_{\Delta t \rightarrow 0} (\Delta t)^{0.5} \frac{N-1}{2} \Delta t \\ &= \lim_{\Delta t \rightarrow 0} (\Delta t)^{0.5} \tau = 0 \end{aligned}$$

Thus, it does not contribute in the limit.

Ito's Lemma

We find the following result, called Ito's lemma:

If $dX = a(X, t)dt + b(X, t)dZ$ and $F = F(X, t)$, then

$$dF = \left(aF_X + \frac{b^2}{2}F_{XX} + F_t \right) dt + bF_X dZ$$

or

$$F(t) - F(0) = \int_0^t \left(aF_X + \frac{b^2}{2}F_{XX} + F_t \right) dt + \int_0^t bF_X dZ$$

This resembles the chain rule but for *stochastic* (random) processes.

So e.g., if we have a stock price X evolving via an SDE, and another variable F that depends on that stock's value, we can use this to determine the SDE describing how F evolves.

No arbitrage lattice pricing

Next, we would like to generalize the no-arbitrage option pricing ideas from our simple two-state tree to more realistic cases, where the stock evolves using a lattice walk/Brownian motion model.

We will (still) consider two basic types of option contracts:

- a **European** style: can only be exercised at maturity T
- b **American** style: can be exercised at any time in $[0, T]$

Stock model – Geometric Brownian Motion

Our underlying lattice method will use a variant of Brownian motion, called Geometric Brownian Motion (GBM), to approximate/model the movement of stock prices. It is defined by the SDE:

$$dS = a dt + b dZ$$

where $a = \mu S$, $b = \sigma S$.

(Equivalently, $dS = \mu S dt + \sigma S dZ$, or $\frac{dS}{S} = \mu dt + \sigma dZ$)

Under this model, S will (at least in theory) remain non-negative, like actual stocks.

Notice that the a, b coefficients explicitly contain S itself.
Let's use Ito's lemma to see if we can simplify this slightly...

Applying Ito's lemma

If X is some function of S , i.e., $X = X(S, t)$, Ito's lemma, in these variables, tells us:

$$dX = \left(a X_S + \frac{b^2}{2} X_{SS} + X_t \right) dt + b X_S dZ$$

Now, we choose $X = X(S, t) = \log(S)$. Then, we can directly differentiate $\log(S)$ to get the derivatives needed above:

$$X_S = \frac{1}{S}, \quad X_{SS} = -\frac{1}{S^2}, \quad X_t = 0$$

Applying Ito's lemma

Plugging those derivatives (and also $a = \mu S$ and $b = \sigma S$) into Ito's lemma, we find:

$$dX = \left(\mu S \left(\frac{1}{S} \right) + \frac{\sigma^2 S^2}{2} \left(-\frac{1}{S^2} \right) + 0 \right) dt + (\sigma S) \frac{1}{S} dz$$
$$dX = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dz$$

This is just another SDE with a random term and a drift term.

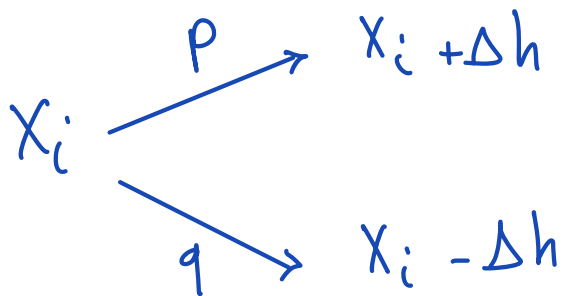
However, this SDE is in a more convenient form.

Thus, in practice, we can work with either variable, S or X , and just use a logarithm $X = \log(S)$ or exponential $S = e^X$, to recover the other one!

Random walks for GBM

Recall we saw that, for a discrete random walk with parameters $\Delta h = \sigma\sqrt{\Delta t}$, $p = \frac{1}{2}(1 + \frac{\alpha}{\sigma}\sqrt{\Delta t})$, $q = 1 - p$, the walk converges in the limit of $\Delta t \rightarrow 0$ to the solution of an SDE of the form:

$$dX = \alpha dt + \sigma dz$$

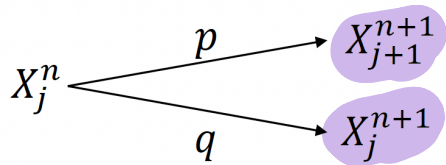


If we choose $\alpha = \mu - \frac{\sigma^2}{2}$, we have the same SDE we just derived from GBM. i.e., a random walk with these parameters will converge to the solution of GBM as $\Delta t \rightarrow 0$.

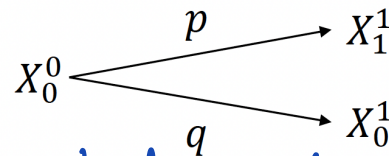
So: We can use the lattice to approximately solve GBM.

Relating X and S in our GBM lattice

Let's label/index our variables on the lattice as follows:



Superscript n indicates timestep, subscript j indicates the vertical position in the lattice.
e.g., on the first step we see

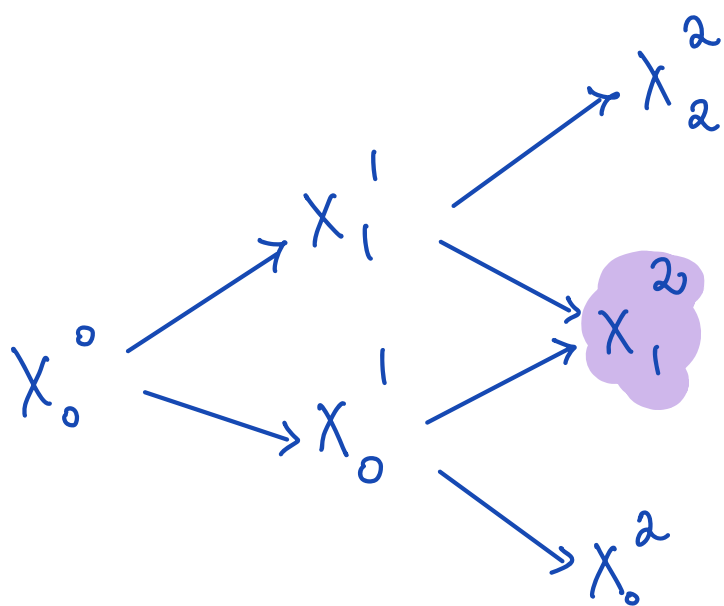


At node (n, j) , you branch to two children at time $n+1$:

- Down move: Subtract $\sigma\sqrt{\Delta t}$. This keeps you at the same vertical index j . So the child is X_j^{n+1} .
- Up move: add $\sigma\sqrt{\Delta t}$. This shifts you to the next higher index.

Recall that $X_j^{n+1} = X_j^n - \sigma\sqrt{\Delta t}$ (down) and $X_{j+1}^{n+1} = X_j^n + \sigma\sqrt{\Delta t}$ (up).
So the child: X_{j+1}^{n+1}

Since we carefully choose $X = \log(S) \rightarrow S = e^X$, lattice node X values can also be replaced using $S_j^n = e^{X_j^n}$.

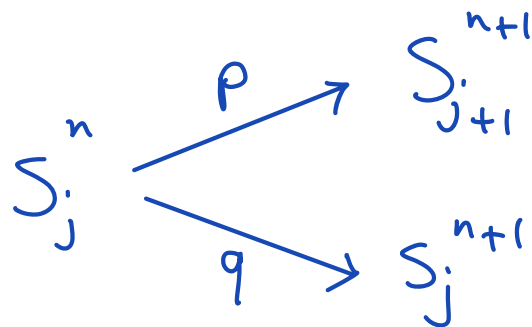


Relating X and S in our GBM lattice

We then get a new lattice, with new up/down update rules in terms of S :

$$S_j^{n+1} = S_j^n e^{-\sigma\sqrt{\Delta t}}$$

$$S_{j+1}^{n+1} = S_j^n e^{+\sigma\sqrt{\Delta t}}$$



So, we have a method to evolve stock prices S on a lattice under GBM. Next, we'll return to the question of how to price options...

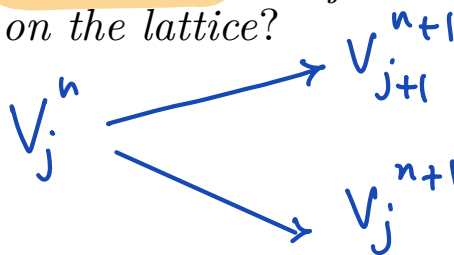
No arbitrage lattice pricing strategy

Similar to our two-state scenario, but extended to the lattice...

- 1 Construct a discrete hedging portfolio.
- 2 Apply the discrete no arbitrage principle.
- 3 Let $\Delta t \rightarrow 0$.
- 4 Get the no arbitrage price and hedging strategy.

Let $V_j^n = V(S_j^n, t^n)$ = the discrete option value at S_j^n on the lattice.

How should the option value evolve on the lattice?

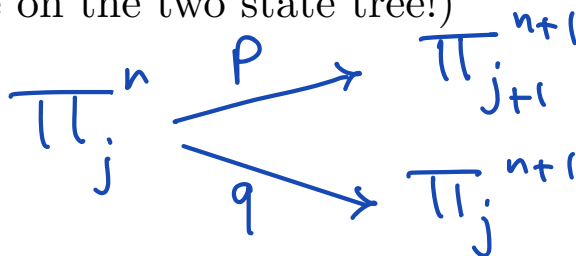


No arbitrage lattice pricing

Construct a hedging portfolio Π at one particular lattice node (n, j) :

$$\pi_j^n = V_j^n - \delta_j^n S_j^n$$

i.e, we have an option with value V and some number δ_j^n of shares in the stock with value S (just like on the two state tree!)



After a time step, two possibilities exist for the new portfolio value:

$$\Pi_j^{n+1} = V_j^{n+1} - \delta_j^n S_j^{n+1}$$

$$\Pi_{j+1}^{n+1} = V_{j+1}^{n+1} - \delta_j^n S_{j+1}^{n+1}$$

No arbitrage lattice pricing

To eliminate risk during the timestep, we choose, δ_j^n to guarantee *no change* in the portfolio value whether we go up or down, i.e., the portfolio value is certain (and risk-free) at $n + 1$:

$$\begin{aligned} \text{Down Value} &= \text{Up Value} \\ V_j^{n+1} - \delta_j^n S_j^{n+1} &= V_{j+1}^{n+1} - \delta_j^n S_{j+1}^{n+1} \end{aligned}$$

which can be solved for δ to get

$$\delta_j^n = \frac{V_{j+1}^{n+1} - V_j^{n+1}}{S_{j+1}^{n+1} - S_j^{n+1}} \approx \frac{\partial V}{\partial S}$$

As $\Delta t \rightarrow 0$, this approximation converges to the derivative $\frac{\partial V}{\partial S}$, also called the option delta.

No arbitrage lattice pricing

Since the portfolio is risk-free, it earns the risk-free rate, r . We can relate the start (n) and end $(n + 1)$ value over the step Δt as:

$$\Pi_j^n = e^{-r\Delta t} \Pi_{j+1}^{n+1} \quad (= e^{-r\Delta t} \Pi_j^{n+1})$$

or

$$V_j^n - \delta_j^n S_j^n = e^{-r\Delta t} (V_{j+1}^{n+1} - \delta_j^n S_{j+1}^{n+1})$$

We already know $S_{j+1}^{n+1} = S_j^n e^{+\sigma\sqrt{\Delta t}}$ and $\delta_j^n = \frac{V_{j+1}^{n+1} - V_j^{n+1}}{S_{j+1}^{n+1} - S_j^{n+1}}$, so we can plug in to get an expression for the time n option value V_j^n in terms of only time $n + 1$ data.

No arbitrage lattice pricing

The result (after more algebra...) is:

$$V_j^n = e^{-r\Delta t} (p^* V_{j+1}^{n+1} + q^* V_j^{n+1})$$

$$u = \frac{1}{d} = e^{\sigma\sqrt{\Delta t}}$$

where we define $p^* = \frac{e^{r\Delta t} - d}{u - d}$ and $q^* = 1 - p^*$.

(The *real* up/down probabilities of the stock do not appear! Risk-free!)

What does the top expression “look” like? Looks like the *expected value* of the quantity V at time $n + 1$, according to the (pseudo-)probabilities p^*, q^* , *discounted* back to time n at interest rate r . Very interesting!

Risk neutral probabilities

For sufficiently small Δt , i.e. $\Delta t \leq \frac{\sigma}{r} \rightarrow r\Delta t \leq \sigma\sqrt{\Delta t}$, we are assured that $0 \leq p^*, q^* \leq 1$.

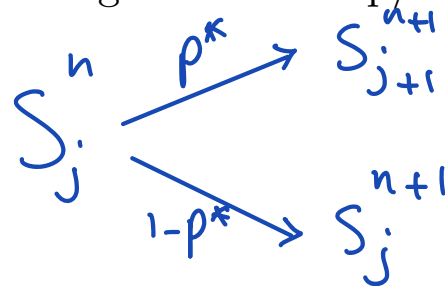
Thus, we can still think of these (very loosely!) as “probabilities”, or rather “risk neutral probabilities”, since we derived them by eliminating risk.

This hints at the idea of what we call a risk neutral random walk.

Risk Neutral Random Walk

Suppose that we now walk on a lattice of this form, using these new up/down “probabilities” p^* , where

$$p^* = \frac{e^{r\Delta t} - d}{u - d} \quad \text{and}$$
$$u = \frac{1}{d} = e^{\sqrt{2}\Delta t}$$



(i.e., we replace the probabilities from slide 13)

We call this the no-arbitrage lattice, and the corresponding walk a risk neutral walk.

Risk Neutral Random Walk

It turns out that, as $\Delta t \rightarrow 0$, the risk-neutral walk on this no-arbitrage lattice converges to the SDE:

$$dS = rSdt + \sigma Sdz$$

i.e., same form as our GBM, but “real world” mean μ is replaced by the *risk-free* rate r of the “risk neutral world”.

Theorem: Under GBM, the no-arbitrage value of an option can be computed by taking the option’s *expected value* in this “risk-neutral world”, assuming assets follow $dS = rSdt + \sigma Sdz$, and *discounting* at the risk-free rate.

Risk neutral world

This is a pretend “world” - *it does not exist*. It is simply a mathematical idea or trick.

The real asset still *actually* follows: $dS = \mu S dt + \sigma S dZ$

We refer to the real world as the P measure.

We refer to the risk neutral world as the Q measure.

Takeaway

Okay, that's all great. But what is the practical outcome?

We can find the S values on the nodes of this no arbitrage lattice, since we have *forward* update formulas for them (slide 8) and we know the *starting* value. This tells us how the stock/asset S behaves in the Q measure (risk-neutral world).

Then, since we also know the option value at the *end* from its payoff, we will be able to walk *backward* through the lattice to determine the discounted expected value (slide 13). This gives the present-day ($t = 0$) no arbitrage option value!

This leads us to specific algorithms for European and American options, which we'll explore next time.

End of Lecture 7